

Alexander Yao

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GitHub LinkedIn

EDUCATION

University of Maryland, College Park
B.S. in Computer Science
Minor in Applied Mathematics

College Park, MD
Expected Dec. 2026

- **Relevant Coursework:** Algorithms, Compilers, Advanced Data Structures, Computer Systems, Programming Languages, Web Development, Parallel Computing, Probability and Statistics, Computer and Network Security

SKILLS

Languages: Python, Java, Rust, C, OCaml, JavaScript, SQL, Racket
Systems & Backend: Linux, Networking, Async I/O, QUIC, Compilers, Data Pipelines
Tools: Git, Pandas, Playwright, Node.js, Microsoft Azure, Android Studio

PROJECTS

Cross-Platform Video Performance Tracker
Python, SQL, Pandas

Dec. 2025

- Designed a cross-platform data ingestion pipeline aggregating engagement metrics from YouTube and TikTok APIs.
- Implemented a headless browser scraping system using Playwright with session persistence and anti-bot mitigation to ensure reliable data collection.
- Developed platform-agnostic engagement normalization metrics (likes + comments per view) to enable fair, data-driven comparison across platforms.
- Transformed raw event data into clean, queryable analytics tables to support performance analysis.

Loot+ Systems Programming Project
Racket, x86-64 Assembly

May 2025

- Implemented a Racket-to-x86-64 compiler supporting closures, variadic functions, and dynamic arity checks.
- Built a custom exception-handling mechanism via stack unwinding and non-local jumps using low-level register manipulation.
- Implemented a full compiler pipeline (parsing, IR generation, codegen), achieving 100% behavioral parity with an interpreted reference model.

EXPERIENCE

George Mason University
Computer Vision & Security Research Assistant

Fairfax, VA
May 2025 – Present

- Designed and implemented a low-latency, cloud-native media delivery system over QUIC, achieving <150ms end-to-end latency.
- Built a scalable bidirectional client-server architecture integrating Python async I/O and Android clients for continuous media streams.
- Improved system reliability by 30% through packet scheduling, flow control tuning, and congestion simulation.
- Developed security testing modules to audit QUIC handshake and encryption behavior, strengthening transport-layer robustness.

Future Scientist Exchange Program
AI Security Research Intern

Hefei, China
June 2025 – August 2025

- Designed and executed controlled experiments simulating instruction-based and clean-label backdoor attacks against large language models.
- Built automated Python evaluation pipelines using statistical analysis to assess robustness and evasion behavior.
- Presented technical findings to an international academic audience of 200+ attendees.

LEADERSHIP

Taiwanese-American Student Association
Junior Officer, Videographer, Senior Advisor

College Park, MD

- Directed and produced high-engagement promotional media for a 300+ member organization, generating 10,000+ views across multi-platform social channels to drive community growth and event attendance.
- Advised an executive board of over 18 members weekly, providing logistical and technical guidance during large event planning.